

Appendix C: Falsifiable Predictions and Observational Constraints

C.1 Overview: Bridging Theory to Observation

The BMI framework is not merely a descriptive model; it is a falsifiable theory. This appendix outlines the three primary observational channels through which the manifold interaction can be verified or refuted.

C.2 CMB B-Mode Spectral Spikes

Standard inflationary models predict a smooth spectrum of B-mode polarization in the Cosmic Microwave Background (CMB). BMI Theory predicts the presence of anomalous spectral spikes in these B-modes. These are resonance-gated features occurring at specific harmonic frequencies corresponding to the “Crystallization Threshold” (10^{-12} s). * **Target Observation:** Simons Observatory / CMB-S4 data. * **Criterion for Verification:** Presence of high-Q resonance peaks in the B-mode power spectrum that deviate from the standard inflationary power-law.

C.3 LVK Gravitational Wave Spectroscopy

BMI Theory requires that binary black hole mergers undergo a dual-mode ring-down. The primary tensor strain is accompanied by a scalar “breathing mode” governed by the manifold coupling coefficient K . * **Target Observation:** LVK (LIGO-Virgo-KAGRA) O4b/O5 strain data. * **Criterion for Verification:** A consistent frequency split ($\Delta f \approx 15$ Hz) observed across distinct detector baseline orientations, excluding instrumental noise floors.

C.4 Collider Cross-Section Anomalies (ILC/CLIC)

In precision electron-positron collider experiments, the theory predicts a systematic energy deficit channel. This is interpreted as energy leakage into the auxiliary shadow manifold via the screening gate (g_{eff}). * **Target Observation:** ILC/CLIC high-luminosity runs. * **Criterion for Verification:** Measurement of a persistent, non-random missing energy signature in the final state of e^+e^- annihilation, correlating with the theoretical coupling constant $K \approx 0.13$.